

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain $\geq 80\%$ accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

6. Retail Motion Tracking Logic

A retail system requires:

- Accurate tracking when crowd density is low
- Stable tracking when density is high

Observations:

Optical Flow: accurate in low density, noisy in high density

Motion Models: stable in high density, less detailed

Compare both and evaluate a suitable approach or combination. Justify your reasoning logically.

9. Mobile Face Detection Trade-off

Constraints:

- Real-time operation required
- Accuracy $\geq 85\%$

Options:

Viola-Jones: 80%, fast

Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

11. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Works under varying lighting

Options:

Method A: 88%, 60 ms

Method B: 91%, 85 ms

Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

Compare and evaluate the suitable model. Justify using constraints.

13. Security System Recognition

A security system must:

- High accuracy ($>92\%$)
- Moderate computation

Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

14. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:

Optical Flow: accurate, slow

Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

17. Industrial Inspection System

Requirements:

- Detect defects with $\geq 95\%$ accuracy
- Real-time operation

Options:

Method A: 92%, fast

Method B: 96%, slow

Method C: 95%, moderate

Compare and evaluate the suitable method. Justify your decision.

29. Autonomous Drone Tracking

Constraints:

- Real-time (<40 ms)
- Accuracy $\geq 85\%$

Options:

Model A: 83%, 30 ms

Model B: 86%, 50 ms

Model C: 88%, 38 ms

Compare and evaluate. Justify your decision.

Code of Conduct:

*I **K.ASWINI(192312397)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

ANSWERS

4. Background Modeling Evaluation

- **Static scenes:** 90%
- **Dynamic scenes:** 65%
- **Required:** $\geq 80\%$ in all cases

Comparison:

- Performs well in static environments
- Fails significantly in dynamic conditions

Evaluation:

- Average performance is inconsistent
- Violates the minimum requirement in dynamic scenes

Conclusion:

GMM alone does NOT meet requirements

Decision: Use **hybrid approach** (e.g., GMM + optical flow / deep learning) to handle dynamic backgrounds

6. Retail Motion Tracking Logic

- **Optical Flow:**
Accurate (low density)
Noisy (high density)
- **Motion Models:**
Stable (high density)
Less detailed

Evaluation:

- Each method performs well under different conditions
- No single method satisfies both requirements

Conclusion:

Use a **hybrid/adaptive system**:

- Low density → Optical Flow
- High density → Motion Models

Ensures both **accuracy + stability**

9. Mobile Face Detection Trade-off

- **Viola-Jones:** 80% (below requirement), fast
- **Deep Learning:** 95% , slow

Constraints:

- Accuracy $\geq 85\%$
- Real-time required

Evaluation:

- Viola-Jones fails accuracy constraint
- Deep learning satisfies accuracy but may violate real-time

Conclusion:

Use **optimized deep learning** (e.g., lightweight CNN, MobileNet)
Meets both constraints with optimization

11. Traffic Surveillance Model Selection

Method Accuracy Latency

A	88%	60 ms
B	91%	85 ms
C	92%	75 ms

Evaluation:

- A fails accuracy
- B fails latency
- C satisfies all constraints

Conclusion:

Method C is the best choice (meets all requirements)

13. Security System Recognition

Approach	Accuracy	Computation
Traditional	85%	Low
Deep Learning	94%	High
Hybrid	91%	Moderate

Evaluation:

- Only deep learning meets accuracy requirement (>92%)
- Computation is high but acceptable for security systems

Conclusion:

Deep Learning is best

Accuracy is critical in security → prioritize it over computation

14. Motion Tracking in Sports Analytics

- **Optical Flow:** Accurate but slow
- **Motion Estimation:** Fast but less accurate

Constraints:

- Accuracy + Real-time required

Evaluation:

- Optical Flow fails real-time

- Motion Estimation fails accuracy

Conclusion:

Use **hybrid approach** (coarse motion estimation + refined optical flow)

Balances speed and accuracy

17. Industrial Inspection System

Method Accuracy Speed

A	92%	Fast
B	96%	Slow
C	95%	Moderate

Evaluation:

- A fails accuracy
- B fails real-time
- C satisfies both

Conclusion:

Method C is the best choice

29. Autonomous Drone Tracking

Model Accuracy Latency

A	83%	30 ms
B	86%	50 ms
C	88%	38 ms

Constraints:

- Accuracy $\geq 85\%$
- Latency < 40 ms

Evaluation:

- A fails accuracy
- B fails latency
- C satisfies both

Conclusion:

Model C is the optimal solution

